

Supplementary Material for “Training Deep Neural Networks to Meet Hard Prediction Quotas”

Anonymous submission

Algorithm S1 TraLO training loop (main paper Sec. 3)

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1: warm up  $\theta$  on  $L_{CE}$  alone; cache the warmup model
2: for each constraint epoch do
3:   update  $\theta$  on  $L_{CE} + \lambda_g \ell_g + \lambda_l \ell_l$  (plus the hinge once
     satisfied)
4:   recount hard predictions on the test inputs
5:   if a cap is violated then
6:     ratchet that cap's  $\lambda$  up one step; advance  $\rho$ 
7:   else if first time all caps are met then
8:     freeze  $\lambda$  and  $\rho$ ; reset the Adam state
9:   end if
10:  stop after 5 consecutive satisfied epochs
11: end for
12: post-hoc verification (flip borderline predictions) if
    needed

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A Mechanism: Bounded vs. Unbounded Dual Behavior

Proof of Lemma 1. Write $u = E/K \geq 0$. In Eq. (1) the first term $E/(E+K) = u/(1+u)$ increases from 0 to 1 and the second $\rho u^2/(1+u^2)$ increases from 0 to ρ , so $0 \leq \ell < 1 + \rho$. Differentiating through $E = \max(0, S-K)$ gives Eq. (2) for $S > K$. There the first summand $K/(E+K)^2 = \frac{1}{K}(1+u)^{-2}$ is at most $1/K$ (at $u=0$) and decreasing; the second, $\frac{2\rho}{K} u/(1+u^2)^2$, is maximized at $u = 1/\sqrt{3}$ with value $\frac{2\rho}{K} \cdot \frac{9}{16\sqrt{3}} = \frac{9\rho}{8\sqrt{3}K} < \rho/K$ and tends to 0 as $u \rightarrow \infty$. Their sum therefore lies in $[0, (1+\rho)/K]$ and vanishes as $E \rightarrow \infty$. Scaling ℓ and $\partial\ell/\partial S$ by a frozen multiplier λ gives the bounds $\lambda(1+\rho)$ and $\lambda(1+\rho)/K$, neither depending on E . $\square$

Proof of Proposition 1. $E_c = \max(0, S_c - K_c)$ is a function of the scalar S_c , and ℓ is a function of E_c , so $\lambda_c \ell(E_c)$ is the composition $\theta \mapsto S_c(\theta) \mapsto \lambda_c \ell$. The chain rule gives $\nabla_\theta[\lambda_c \ell] = \lambda_c \ell'(E_c) (\partial E_c / \partial S_c) \nabla_\theta S_c = \lambda_c (\partial \ell / \partial S_c) \nabla_\theta S_c$, using $\partial E_c / \partial S_c = \mathbf{1}[S_c > K_c]$. Set $a_c = \lambda_c \partial \ell / \partial S_c$, which is ≥ 0 by Lemma 1. Summing over caps gives a gradient in $\text{span}\{\nabla_\theta S_c\}_c$, a subspace of dimension at most the number of caps. $\square$

Proof of Corollary 1. Let $g = \nabla_\theta L_{CE} + \sum_{c'} a_{c'} \nabla_\theta S_{c'}$. In the limit where one cap's term dominates and the re-

maining terms vanish, $g \rightarrow a_c \nabla_\theta S_c$ with $a_c > 0$; norm-clipping, when active ($\|g\| > \tau$), then returns $\tau g / \|g\| \rightarrow \tau \nabla_\theta S_c / \|\nabla_\theta S_c\|$, in which the positive scalar a_c (hence λ_c , ρ , and the shape of ℓ) cancels. $\square$

Proof of Proposition 2. Write $g(S) = \ell(\max(0, S-K); K, \rho)$. For $S < K$, $g \equiv 0$, so g has left-derivative 0 at K . For $S > K$, $g'(S) = K/(E+K)^2 + \frac{2\rho}{K} u/(1+u^2)^2$ with $E = S-K$ and $u = E/K$ (Eq. (2)); as $S \rightarrow K^+$ the first term $\rightarrow 1/K$ and the second $\rightarrow 0$, so g has right-derivative $1/K$ at K , independent of ρ . Parameterize a path along which the soft count rises through K , and let $c = |dL_{CE}/dS|_{S=K}$ be the cross-entropy pull at the cap; the constrained class is under-served at a cap $L < 100\%$, so $dL_{CE}/dS \leq 0$ there. The one-sided derivatives of $F = L_{CE} + \lambda g$ at K are $F'(K^-) = -c \leq 0$ and $F'(K^+) = -c + \lambda/K$. If $\lambda \geq cK$ then $F'(K^+) \geq 0$, so F does not decrease past the cap: $S=K$ is a local minimum along the path, and a finite multiplier attains the cap. The nonzero boundary slope $1/K$ is what makes the penalty exact, playing the role of the ℓ_1 penalty's unit slope (Nocedal and Wright 2006). $\square$

Figure S1 plots these two bounds against the violation size. Why does a hard count cap stress the dual-ascent baselines? On the paper grid the interaction is hidden: after the 50-epoch warmup the classifier is already saturated, and a saturation gate shared by every method (train accuracy ≥ 0.995 on two consecutive checks) stops cross-entropy updates within the first two constraint epochs, so the constraint phase is pure count optimization for all methods alike. We therefore probe with a one-epoch warmup, where the classifier keeps learning throughout the constraint phase and the two objectives visibly compete (Figure S2).

The probe shows three things. First, while the classifier is learning, the count cannot be forced down: each epoch's cross-entropy updates re-inflate the constrained class, and the excess oscillates around its starting level for ~ 60 epochs under both methods (Fig. S2c). Dual ascent responds exactly as designed: the violation persists, so the multiplier grows every epoch, from 1 to 53 (seeds 2-3: 58, 60). In control terms this is integrator windup: the controller keeps integrating an error it has no ability to act on. Second, the escalation buys nothing. Every method takes a single gradient-clipped constraint step per epoch (max-norm 1.0, shared across methods), so extra

multiplier cannot buy extra descent: the two cross-entropy traces are numerically indistinguishable (Fig. S2b), and the count comes down no faster at $\lambda=53$ than at TraLO’s ratcheted $\lambda \leq 0.19$. The asymmetry is in what the clip rail is doing. At $\lambda=53$ an unbounded-penalty gradient would dominate the update by orders of magnitude were it not clipped, whereas TraLO’s bounded penalty at $\lambda \leq 0.19$ demands orders of magnitude less force in the first place. Third, the count falls only once the classifier saturates and the gate stops CE updates (dotted lines in Fig. S2c), and then both methods reach the cap within a few epochs. The count outcome is identical; what differs is the pressure applied to the constrained class to get there.

The paper grid shows the same behavior in compressed form: the dual multipliers escalate while TraLO’s freezes (≤ 0.09 there), Fioretto’s native satisfaction is ~ 0.87 (0.84 on OctMNIST and TissueMNIST, 0.93 on DermMNIST), so its unbounded multiplier stalls short of the cap on $\sim 13\%$ of runs that never reach feasibility within the budget (main paper Sec. 5.4); runs that end violated restore their best-satisfied checkpoint and remain competitive.

Nor is the escalation a step-size artifact. A dedicated fairness sweep (Block C of the released corpus; 96 runs on the hard-regime cells) varies Fioretto’s dual step over 0.001–0.01 and Hounie’s η over 0.001–0.1: Fioretto’s constrained-class F1 moves by less than 0.001 across the $10\times$ step range, larger Hounie steps only hurt (0.43 $\rightarrow$ 0.35), and no setting closes the gap to TraLO on the OctMNIST tight cells (the best variant still trails by 0.019 cc-F1).

We state this mechanism *qualitatively only*. An audit shows that the magnitude of the escalation does not predict where TraLO wins (the multiplier is in fact largest in some cells where TraLO loses). So the claim is “dual ascent needs an unbounded multiplier plus a clip rail to hold a hard count, while TraLO’s bounded, frozen multiplier holds it alone,” not any dose–response of the form “more escalation $\Rightarrow$ bigger win.”

B Complete Component Ablation

A within-cell ablation isolates which ingredients carry the hard-regime advantage: on the OctMNIST tight-cap cells where TraLO leads, we disable one component at a time and measure the paired change in constrained-class F1 (Table S1; $n=24$ per component). Two components are load-bearing. Removing the optimizer reset at satisfaction costs $+0.079$ cc-F1, the single largest effect, helping on all 24 paired runs and hence in all six cells (cell-level sign test $p=0.031$; we test at the cell level because the 24 runs share seeds across cells), because it breaks the post-satisfaction descent momentum that would otherwise erode the constrained class. Removing the undershoot hinge (the TraLO-bounded control, already in the frozen grid) costs $+0.036$ cc-F1 (positive on all 24 paired runs; cell-level sign test $p=0.031$; 0.019–0.070 across cells). The other components are cc-F1-neutral on these already-satisfied cells: the ρ schedule and the lambda freeze move it by under 0.001 (n.s.). A dedicated KL sweep on TissueMNIST (the $L30$ tight cap, MobileNetV3, $\alpha_{KL} \in \{0, 0.1, 0.3, 1.0\}$) confirms the anchor is off for good reason: $\alpha_{KL}=0$ (the paper recipe) gives the best

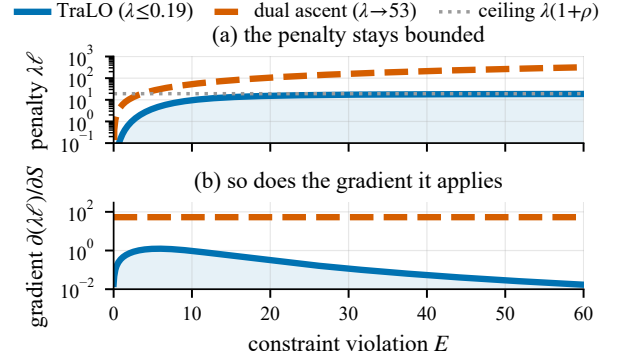


Figure S1: **The bound of Lemma 1, drawn.** At the multipliers each method reaches on the one-epoch probe, TraLO’s per-cap penalty (a) flattens at its ceiling $\lambda(1+\rho)$ and the gradient it applies (b) stays capped and decays far from the cap, while dual ascent’s linear penalty and constant $\lambda \approx 53$ gradient grow without bound in E and λ . This is the shape of the *objective*; whether the optimizer is sensitive to it is the empirical question the body answers: the scale is rescaled away, and the shape proves neutral by ablation and graft (Sec. B). Analytic, $K=10$.

constrained-class *and* macro-F1, while every nonzero anchor lowers cc-F1 (by 0.01–0.03) and only adds prediction edits. The KL term was a useful drift damper in an early version of the method, before the optimizer reset and undershoot hinge; once those two control post-satisfaction drift directly, the anchor is redundant. The advantage rests on the reset and the hinge. One caveat applies to all rows: the component importances are estimated inside the selected win region (the tight-cap cells where TraLO leads), so they quantify what carries that win, not what the components do elsewhere on the grid. The reset, ρ , and freeze legs are a dedicated ablation under the identical frozen recipe; the hinge leg is already in the main grid.

B.1 Grafting the two components onto the dual baselines

The ablation above removes components from TraLO; the converse test adds them to the baselines. We graft exactly the optimizer reset and the undershoot hinge (each using the host’s own multipliers and penalty) onto Fioretto-LDF and Hounie-RCL and rerun the six tight-cap cells: 2 caps $\times$ 3 backbones $\times$ 4 seeds, 144 runs in all, including re-run controls for TraLO and both hosts under identical warmup caches, the frozen recipe, and one GPU model. The re-run controls reproduce the main-grid values to ± 0.0013 in all 18 control cells, so the arms are compared on equal footing. Table S2 reports mean cc-F1 per cell.

The graft lifts Fioretto-LDF in six of six cells ($+0.019$ to $+0.073$ cc-F1) and Hounie-RCL in five of six ($+0.035$ to $+0.103$; the one exception, $L40$ on RegNet, drops by 0.028), recovering a median 86% of TraLO’s margin over the Fioretto host and 92% over the Hounie host (medians over the six

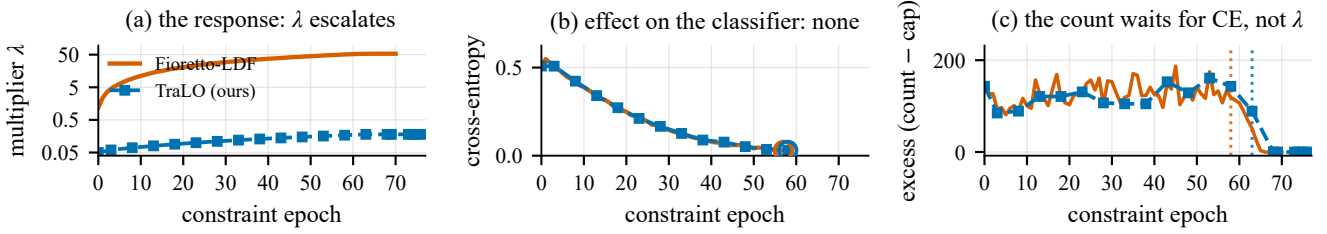


Figure S2: **The escalating multiplier is discarded, not used: a one-epoch-warmup probe** (DermMNIST $L50/G50$, RegNetY-400MF, seed 1), the empirical counterpart to Prop. 1. (a) Dual ascent integrates the standing violation into a multiplier that climbs to $\lambda=53$, while TraLO’s ratchet freezes at $\lambda \leq 0.19$. (b) The effect on the classifier is none: every method takes a single norm-clipped constraint step per epoch, so the two cross-entropy traces are numerically indistinguishable. (c) The count does not fall until cross-entropy finishes (train-accuracy gate, dotted), then reaches the cap for both. A $\sim 280\times$ gap in λ thus leaves training nearly identical, consistent with Cor. 1: with the constraint term dominant and the step norm-clipped, the update is independent of λ . Adam’s invariance to a constant rescaling of the gradient (Kingma and Ba 2015) points the same way, though we read *that* as an empirical observation rather than a theorem, since λ multiplies only the constraint term (so it shifts the cross-entropy-versus-count weighting) and the Adam cancellation is exact only as $\epsilon \rightarrow 0$ and where the constraint term dominates.

Table S1: **Complete component ablation on the OctMNIST hard-binding cells.** Effect on constrained-class F1 of removing each TraLO component in turn ($n=24$ paired runs each: 2 tight caps $\times$ 3 backbones $\times$ 4 seeds). Δ cc-F1 = full TraLO – (component removed); winrate = fraction of the 24 paired runs where removing the component hurts. Inference is at the cell level (the unit that does not repeat seeds): removing either load-bearing component hurts in all six cells (sign test $p=0.031$, the six-cell floor), while the ρ schedule and lambda freeze split the cells and are cc-F1-neutral on these already-satisfied cells. Two essential ingredients plus two neutral ones, *not* four co-equal components.

Component removed	Δ cc-F1	Winrate
<i>Load-bearing</i>		
Optimizer reset	+0.079	1.00
Undershoot hinge	+0.036	1.00
<i>Neutral</i>		
ρ schedule	+0.001	0.42
Lambda freeze	+0.000	0.38

cells, the loss cell included). Against each grafted host separately TraLO stays best-or-tied (8 wins, 4 ties, 0 losses in the twelve paired comparisons under the main-text win rule). Taken seed-by-seed against the stronger of the two grafted hosts, though, its mean edge is only +0.0015 and it trails in two of the six cells: once the recipe is grafted the three methods are statistically interchangeable, with what remains of TraLO’s lead concentrated on the ViT-B/16 (+0.015/+0.017 over the grafted Fioretto-LDF at the two caps). On macro-F1, TraLO versus the grafted Fioretto-LDF is a tie in all six cells.

As an anti-windup control, we also run Fioretto-LDF with dual restarts: all multipliers are reset to zero the moment the soft counts are feasible, following Gallego-Posada et al. (2022). Its deployed predictions are *identical* to plain Fioretto-LDF in every cell, although every run fires at least

one restart. The reason is mechanistic: the deployed checkpoint is selected on the constraint-excess axis before first feasibility is reached, and up to that epoch the two arms are step-for-step identical, so machinery that only alters post-feasibility dynamics cannot change the deployed model. TraLO beats this arm in all six cells. The dual baselines’ tight-cap deficit is therefore not an artifact of missing anti-windup machinery, and the linear-penalty comparison in the main grid is not a strawman in that sense.

B.2 The penalty shape is neutral

The graft removes the question of whether TraLO’s *host* matters; a third arm removes the question of whether its *penalty* does. We rerun TraLO on the same six cells with Eq. (1)’s saturating penalty replaced by a linear one, $\ell = E/K$, holding the ratchet, the freeze, the optimizer reset and the undershoot hinge fixed, so the only change is the shape the method is named for. The arm is genuinely different, and by a wide margin: its logged constraint loss runs about $3\times$ smaller throughout. Its results are not: cc-F1 moves by -0.001 (the bounded shape wins 9 of 24 runs and loses 9, sign-test $p=1.0$; the six cell means split three positive and three negative, none beyond ± 0.006) and median time-to-feasibility by two constraint epochs. In 6 of the 24 runs the two arms land on *identical* cc-F1: changing the penalty shape changed no prediction at all, which the fixed-direction argument below anticipates.

This is what the analysis of Sec. 3 predicts, though the reason is structural rather than a generic property of Adam. The constraint penalty depends on the parameters only through the scalar soft-count S_c , so its gradient is $\ell'(S_c) \nabla_{\theta} S_c$: a scalar times a fixed direction. Multiplying the penalty by a positive constant scales only that scalar, and behind Adam’s coordinate-wise normalization $g \rightarrow cg$ sends $m \rightarrow cm$, $\sqrt{v} \rightarrow c\sqrt{v}$, leaving $\hat{m}/(\sqrt{\hat{v}} + \epsilon)$ unchanged but for the ϵ term; the update along that direction is thus nearly scale-free. The multiplier is such a scalar, so an escalating one

cannot by itself reach the weights: in the escalation probe of Sec. A, where $\lambda=53$ (nearly $280\times$ the probe’s 0.19) drives $\|g\|>1$, it is the max-norm rail that caps the step, and the count falls no faster than at $\lambda \leq 0.19$. The penalty *shape* and the weight ρ reshape $\ell'(S_c)$ rather than rescale it, so they are not covered by the constant- c argument; we establish their neutrality empirically instead, by graft (shape) and by the ablation rows of Table S1 (ρ schedule, freeze). What no rescaling touches is Adam’s own accumulated state, and the hinge, which reverses the direction to $-\nabla_{\theta} S_c$ while the count undershoots: these are the components the ablation ranks first and the graft transfers.

We report this as a negative result about our own formulation. Eq. (1) earns its place by keeping the multiplier small and readable, and by making the two-phase argument of Sec. 3 available at all; it does not earn the constrained-class gains, which belong to the reset and the hinge.

B.3 TraLO’s own hyperparameters are not doing the work

The dual baselines’ step size is swept an order of magnitude to confirm their tuning is not what wins or loses the comparison (Sec. A); fairness asks the same of TraLO’s own continuous knobs, the more so in a transductive setting with no held-out split. On the six OctMNIST tight cells we sweep the hinge weight $\beta \in \{0.25, 0.5, 0.75, 1.0\}$ and the ratchet step $\lambda_{\text{step}} \in \{0.001, 0.002, 0.004\}$ around the frozen recipe ($\beta=0.5$, $\lambda_{\text{step}}=0.002$), one knob at a time, 24 runs per arm, with everything else, warmup cache included, held fixed (120 runs).

The ratchet step is neutral. Moving λ_{step} over a factor of four shifts mean cc-F1 by at most 0.0012 and the median time-to-feasibility by under an epoch. This is the scalar-rescaling argument of Sec. 3 made good: λ_{step} sets only how fast the multiplier climbs, and the multiplier is a positive scalar on the fixed soft-count direction, which Adam’s normalization and the clip rail absorb.

The hinge weight matters, but the win is not a knife-edge at $\beta=0.5$. As the one continuous knob on a load-bearing component, β does move cc-F1, but by at most 0.003 in the six-cell mean across the whole $\{0.25, \dots, 1.0\}$ range, smallest at the weakest hinge $\beta=0.25$, exactly where less hinge restores less constrained-class recall. More to the point, TraLO clears the best trained baseline in all 30 arm-by-cell comparisons, with margins from +0.014 to +0.088: the tight-cap win survives every setting of the two knobs, not only the values we fixed a priori.

B.4 The convergence ordering is not an initialization artifact

TraLO starts its multiplier at $\lambda=0.05$ while the dual baselines start at $\lambda=0$ and integrate upward, so one could read TraLO’s faster feasibility (Sec. 5.4) as a head start rather than a property of the method. We test this on the six OctMNIST tight cells by rerunning each dual baseline from TraLO’s own $\lambda=0.05$ start, changing nothing else (24 runs each). The nonzero start leaves Fioretto-LDF’s median time-to-feasibility unchanged (28 constraint epochs, taken over the

Cap	Backbone	Fioretto-LDF			Hounie-RCL		
		host	+R+H	+restart	host	+R+H	TraLO
L30	MNetV3	0.391	0.412	0.391	0.326	0.411	0.407
	RegNet	0.372	0.399	0.372	0.299	0.402	0.407
	ViT-B/16	0.335	0.409	0.335	0.332	0.416	0.423
L40	MNetV3	0.487	0.510	0.487	0.456	0.491	0.509
	RegNet	0.452	0.470	0.452	0.410	0.383	0.473
	ViT-B/16	0.462	0.496	0.462	0.431	0.506	0.513

Table S2: **Graft and anti-windup arms on the OctMNIST tight caps** (mean cc-F1 over seeds 1–4; **bold** = best in row). +R+H grafts TraLO’s optimizer reset and undershoot hinge onto the host; +restart resets all dual multipliers to zero at soft feasibility (Gallego-Posada et al. 2022). All 144 runs share the frozen recipe, warmup caches, and GPU model with the re-run hosts; the re-run hosts reproduce the main-grid values to ± 0.0013 .

runs that reach feasibility) and it still fails to reach the count natively on 6 of 24 runs; it narrows Hounie-RCL’s median from 40 to 31 (Hounie-RCL reaches feasibility on every run, so its median is over all 24). Both still trail TraLO’s 25. So the head start explains part of Hounie’s gap and little of Fioretto’s, and closes neither: on these cells the ratchet-and-freeze schedule reaches and holds the count faster than either a zero- or a nonzero-start dual ascent. This control is specific to the OctMNIST tight cells; the cross-dataset census of Sec. 5.4 uses each method as published.

C Native Satisfaction: a Deployment Property, not a Headline

We separate *native satisfaction* (meeting the count with no post-hoc edit) from the quality results in the main paper because, on its own, it does not discriminate methods: every deployable system meets a hard count one way or another, and post-hoc clipping reaches satisfaction 1.00 by construction, simply by editing predictions. We report it only to document the mechanism behind TraLO’s quality advantage over clipping (main paper Sec. 5.3): TraLO reaches the count during training and so never invokes the editing step that costs clipping its accuracy. Table S3 gives the deployment costs at the tight L30 cap on MobileNetV3: macro-F1, accuracy, and the number of predictions each method must overwrite (Flips). Figure S3 shows native satisfaction per method over the full grid. TraLO and Hounie-RCL meet the count in every run; Fioretto-LDF and the hinge-free TraLO-bounded do most of the time; the post-hoc clippers meet it natively only at loose caps. A caveat on *flips*: at near-trivial loose caps (e.g. OctMNIST L80/L90) TraLO can record more flips than a clipper that barely has to act, so the robust statement is “satisfies natively while preserving macro-F1,” never a raw flip count.

The macro-F1 advantage over clipping is not specific to MobileNetV3 or to the single L30 cap of Table S3. Table S4 averages the paired gap over the symmetric caps for each dataset–backbone cell, adding MobileNetV2 to the three main-grid backbones: the mean gain is positive in every one

Table S3: **Model quality at the tight $L30$ cap** (MobileNetV3; each value is the mean over seeds 1–4 *at this single cap level*). All methods meet the count; the post-hoc clippers do so by *editing predictions*, and Flips is the number of test predictions overwritten to enforce the cap. This editing erodes **macro-F1**, the imbalance-aware quality metric, on every dataset. Raw *accuracy* is nearly blind to it: the constrained class is rare, so clipping it barely moves the aggregate and the clippers can tie or even beat the trained methods on accuracy while losing macro-F1. This is why F1, not accuracy, is the right quality metric here. [†] = post-hoc clipping. **Bold** = best, underline = 2nd-best per column within each dataset block.

Method	Tissue			Derm			OctMNIST		
	macro-F1 $\uparrow$	Acc. $\uparrow$	Flips $\downarrow$	macro-F1 $\uparrow$	Acc. $\uparrow$	Flips $\downarrow$	macro-F1 $\uparrow$	Acc. $\uparrow$	Flips $\downarrow$
Heur. [†]	0.347	0.478	99.5	0.722	0.846	136.5	0.666	0.695	109.5
Shifman [†]	0.347	0.478	99.5	0.721	0.846	136.5	0.665	0.695	109.5
Fioretto	0.405	0.536	7.8	0.723	0.842	10.2	<u>0.701</u>	0.738	<u>0.2</u>
Hounie	<u>0.406</u>	<u>0.537</u>	11.2	0.728	0.843	19.2	0.678	0.723	0.0
TraLO-b	0.405	0.536	10.0	<u>0.727</u>	<u>0.844</u>	<u>9.8</u>	0.700	0.738	0.0
TraLO	0.408	0.539	7.0	<u>0.726</u>	<u>0.844</u>	1.0	0.702	<u>0.736</u>	0.0

Table S4: **TraLO’s macro-F1 advantage over post-hoc clipping is backbone- and dataset-general.** Mean paired macro-F1 gap of TraLO over the best post-hoc clipper (per-seed best of Heuristic/Shifman-LP), averaged over the symmetric caps for each dataset–backbone cell; *win/tie/loss* counts the caps. The mean gain is positive in every cell, largest on TissueMNIST; it also holds under asymmetric caps (App. D).

Backbone	Tissue	Derm	OctMNIST
MobileNetV3	+0.058 9/0/0	+0.009 6/2/1	+0.044 9/0/0
RegNetY-400MF	+0.082 9/0/0	+0.005 6/2/1	+0.011 6/3/0
ViT-B/16	+0.020 9/0/0	+0.033 9/0/0	+0.007 5/4/0
MobileNetV2	+0.049 6/1/0	+0.004 3/3/0	+0.040 5/0/0

of the twelve cells, largest on TissueMNIST, and it persists under asymmetric caps (App. D).

D Per-Cell Results Across the Full Grid

Tables S8–S11 give the fully granular breakdown behind the regime summary of the main paper (Sec. 5.2): one row per (dataset, backbone, cap), averaged over the four seeds *only*, with no pooling across backbones or caps. Each row pairs TraLO against the best constraint-trained baseline on constrained-class F1 (per-seed best of Fioretto-LDF/Hounie-RCL) and against the best post-hoc clipper on macro-F1, both paired by seed.

Two patterns read straight off the cells. The constrained-class win is *consistent* (all three backbones, every seed) only at the OctMNIST tight caps ($L30/L40$), where it is largest on ViT-B/16; a handful of isolated single-backbone cells elsewhere also clear the +0.005 bar, but no other regime wins across backbones, and none is a loss beyond seed noise. The macro-F1 margin over clipping is positive in 77 of the 81 symmetric cells; the four exceptions are the very tightest $L10/L20$ caps, where the cap nearly zeroes out the constrained class and the gap is within ± 0.02 . The asymmetric caps (Table S11) hold the *preliminary* constrained-class win, on DermMNIST where the global cap is tighter than the group cap. That per-cell table is MobileNetV3, the only back-

bone with the full asymmetric grid; a cross-backbone check on the four shared $G < L$ cap pairs (Table S7) shows the constrained-class win reproduces on RegNet (+0.9pp, 75% seed-winrate) and ties on the ViT (−0.0pp), while the macro-F1 advantage over clipping holds on all three backbones. We report the asymmetric result as preliminary because only MobileNetV3 has been run on the complete asymmetric grid, not because a single backbone shows it.

Table S5 summarizes the constrained-class picture across backbones, adding MobileNetV2 to the three main-grid backbones and averaging the paired gap over the symmetric caps: on TissueMNIST and DermMNIST no backbone shows a systematic advantage (means within ± 0.003), while OctMNIST is positive on all four. The honest tie and the OctMNIST win are each backbone-general, not artifacts of one architecture. Table S6 isolates the OctMNIST tight caps ($L30/L40$) behind that positive row: the paired delta over the best trained dual is positive at every backbone $\times$ cap cell on cc-F1 (all four seeds) and improves macro-F1 at five of the six cells, peaking on the ViT-B/16. This is the per-backbone breakdown the main paper cites next to its consolidated grid (main paper Table 1).

E Datasets and Group Construction

The constrained class is a genuine minority on two of the three datasets and not on the third: GE is 7.1% of the TissueMNIST test set (171/2400) and melanoma is 11.1% of DermMNIST (223/2003), while OctMNIST’s test split is class-balanced by construction (250 per class), making drusen 25%. We keep OctMNIST because it is where the cap binds hardest, not because it matches the screening story: at $L30$ the cap admits 75 of 250 true drusen, so a structural majority of them cannot be flagged whatever the model does. Readers should not take the OctMNIST result as evidence about rare-class screening, and a clinical deployment would more plausibly cap the urgent classes (CNV, DME) than drusen.

Local caps are only over real groups on DermMNIST. There we pool HAM10000’s *localization* field into three anatomical groups: torso (back, trunk, abdomen, chest,

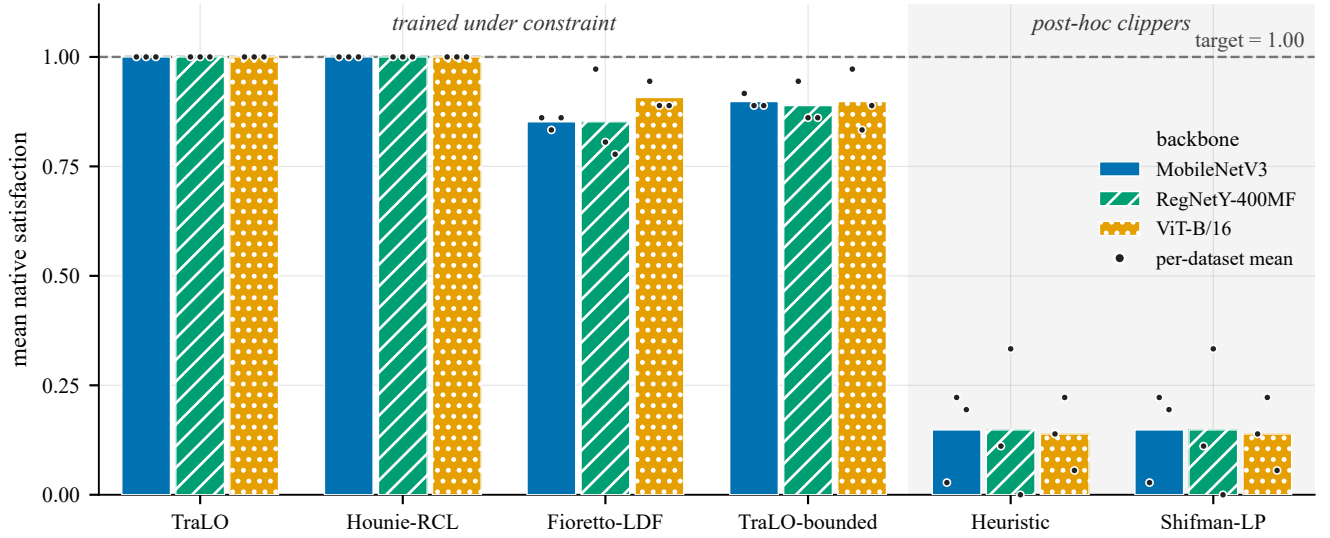


Figure S3: **Methods trained under the constraint satisfy the count natively; post-hoc clippers do not.** Mean native satisfaction per method and backbone over the paper_final grid ($N=108$ runs per method $\times$ backbone: 3 datasets $\times$ 9 caps $\times$ 4 seeds); the dots are the three per-dataset means. TraLO and Hounie-RCL meet the count in every run (satisfaction 1.00); Fioretto-LDF and TraLO-bounded reach it most of the time (~ 0.85 – 0.91); the post-hoc clippers satisfy natively only at loose caps (~ 0.14 – 0.15) and otherwise reach the cap solely by editing predictions, the source of their accuracy tax (Table S3).

Table S5: **The constrained-class picture is backbone-general.** Mean paired constrained-class-F1 gap of TraLO over the best constraint-trained baseline (per-seed best of Fioretto-LDF/Hounie-RCL), averaged over the symmetric caps for each dataset–backbone cell; win/tie/loss counts the caps (a win needs the mean gap and at least half the seeds to clear $+0.005$; loss is the mirror). Across all four backbones (the three main-grid ones plus MobileNetV2), TissueMNIST and DermMNIST show no systematic constrained-class advantage (per-backbone means within ± 0.003 , mostly ties), while OctMNIST is a positive margin on every backbone. The honest tie and the OctMNIST win are both properties of the regime, not of one architecture.

Backbone	Tissue	Derm	OctMNIST
MobileNetV3	-0.002 0/6/3	$+0.002$ 0/9/0	$+0.004$ 2/6/1
RegNetY-400MF	-0.002 1/5/3	-0.003 0/7/2	$+0.011$ 4/5/0
ViT-B/16	-0.001 1/7/1	-0.002 0/7/2	$+0.018$ 5/4/0
MobileNetV2	$+0.002$ 3/3/1	-0.002 1/3/2	$+0.002$ 1/3/1

genital, unknown), extremity (upper/lower extremity, foot, hand, acral), and head/neck (face, neck, scalp, ear), giving 1030/738/235 test images carrying 98/85/40 melanomas. The groups are deliberately unequal in both size and prevalence, which is what makes the per-group caps bind differently.

TissueMNIST and OctMNIST carry no group metadata, so their local caps use a fixed synthetic binary split drawn once from a seeded generator and stored with the data. These groups are arbitrary by construction. They exercise the mech-

Table S6: **The OctMNIST tight-cap win is backbone-general, largest on the ViT-B/16, and carries to overall accuracy.** Paired gap of TraLO over the best trained dual (per-seed best of Fioretto-LDF/Hounie-RCL) at the tight caps, mean over seeds 1–4 ($\pm$ across-seed std). Constrained-class F1 leads at every cell on all 4/4 seeds; macro-F1 improves at five of the six cells (the exception a wash at MobileNetV3 $\times$ L30), again peaking at the ViT-B/16.

Backbone	Cap	Δ cc-F1	Δ macro-F1
MobileNetV3	L30	$+0.016 \pm .014$	$+0.001 \pm .001$
	L40	$+0.022 \pm .029$	$+0.008 \pm .013$
RegNetY-400MF	L30	$+0.035 \pm .006$	$+0.013 \pm .005$
	L40	$+0.021 \pm .003$	$+0.007 \pm .002$
ViT-B/16	L30	$+0.081 \pm .045$	$+0.028 \pm .014$
	L40	$+0.051 \pm .019$	$+0.016 \pm .008$

anism, several simultaneous caps that must hold at once alongside the global cap, and nothing else: no fairness reading of the TissueMNIST or OctMNIST local caps is available, and we do not offer one. Even on DermMNIST, where the groups are real, a local cap is a per-group *ceiling* on flagging melanoma rather than a floor on detecting it, so satisfying it is not evidence of equitable treatment. A deployment that set these caps from estimated prevalence would inherit whatever under-diagnosis those estimates already encode. We report the mechanism and leave the allocation question, which is a question about who sets K_g and on what evidence, outside our claims.

Table S7: **The DermMNIST global-tighter ($G < L$) win reproduces across backbones.** Cross-backbone check on the four shared $G < L$ cap pairs ($L50/G20$, $L50/G30$, $L70/G50$, $L80/G70$; mean over seeds 1–4, $n=16$ paired seed-cells). Δ_{cc-F1} is TraLO minus the best constraint-trained baseline; $\Delta_{macro-F1}$ is TraLO minus the best post-hoc clipper; both paired by seed, with *win* the seed-cell fraction favoring TraLO. The constrained-class win holds on MobileNetV3 and RegNet and ties on the ViT; the macro-F1 advantage over clipping holds on all three.

Backbone	Δ_{cc-F1} vs. trained		$\Delta_{macro-F1}$ vs. clipper	
	Δ	win	Δ	win
MobileNetV3	+0.012	81%	+0.015	77%
RegNetY-400MF	+0.009	75%	+0.002	50%
ViT-B/16	−0.000	38%	+0.032	100%

Table S8: **TissueMNIST: per-cell constrained-class and overall F1.** One row per (backbone, cap), mean over seeds 1–4. cc-F1 columns compare TraLO to the best constraint-trained baseline (per-seed best of Fioretto-LDF/Hounie-RCL, paired); *win* is the seed-winrate. Δ_{mac} is TraLO minus the best post-hoc clipper on macro-F1. **Bold** Δ marks a paired win ($\Delta \geq 0.005$ on $\geq 3/4$ seeds).

Cap	cc-F1 vs. best trained				macro	
	best	TraLO	Δ	win	Δ_{mac}	
<i>MobileNetV3</i>						
L10	0.106	0.101	-0.005	1/4	+0.028	
L20	0.205	0.205	-0.000	1/4	+0.054	
L30	0.279	0.275	-0.005	1/4	+0.061	
L40	0.301	0.300	-0.001	2/4	+0.060	
L50	0.352	0.354	+0.002	2/4	+0.060	
L60	0.381	0.375	-0.006	0/4	+0.064	
L70	0.397	0.399	+0.002	2/4	+0.064	
L80	0.414	0.407	-0.006	1/4	+0.065	
L90	0.409	0.409	+0.000	1/4	+0.065	
<i>RegNetY-400MF</i>						
L10	0.109	0.112	+0.003	1/4	+0.061	
L20	0.212	0.195	-0.017	0/4	+0.073	
L30	0.291	0.273	-0.018	0/4	+0.080	
L40	0.314	0.324	+0.010	2/4	+0.091	
L50	0.362	0.366	+0.004	3/4	+0.085	
L60	0.379	0.383	+0.004	2/4	+0.084	
L70	0.414	0.409	-0.005	1/4	+0.091	
L80	0.429	0.432	+0.003	2/4	+0.089	
L90	0.435	0.431	-0.005	1/4	+0.087	
<i>ViT-B/16</i>						
L10	0.133	0.122	-0.011	0/4	+0.011	
L20	0.239	0.239	+0.000	0/4	+0.016	
L30	0.327	0.327	+0.000	0/4	+0.018	
L40	0.360	0.373	+0.013	4/4	+0.020	
L50	0.422	0.418	-0.004	0/4	+0.024	
L60	0.445	0.447	+0.002	1/4	+0.023	
L70	0.466	0.462	-0.003	1/4	+0.021	
L80	0.487	0.482	-0.005	2/4	+0.025	
L90	0.495	0.497	+0.002	2/4	+0.025	

Table S9: **DermMNIST: per-cell constrained-class and overall F1.** One row per (backbone, cap), mean over seeds 1–4. cc-F1 columns compare TraLO to the best constraint-trained baseline (per-seed best of Fioretto-LDF/Hounie-RCL, paired); *win* is the seed-winrate. Δ_{mac} is TraLO minus the best post-hoc clipper on macro-F1. **Bold** Δ marks a paired win ($\Delta \geq 0.005$ on $\geq 3/4$ seeds).

Cap	cc-F1 vs. best trained				macro	
	best	TraLO	Δ	win	Δ_{mac}	
<i>MobileNetV3</i>						
L10	0.169	0.169	-0.000	1/4	-0.018	
L20	0.310	0.312	+0.002	1/4	-0.001	
L30	0.417	0.419	+0.002	1/4	+0.004	
L40	0.506	0.510	+0.003	2/4	+0.012	
L50	0.563	0.564	+0.002	1/4	+0.014	
L60	0.618	0.618	-0.000	2/4	+0.013	
L70	0.645	0.648	+0.003	3/4	+0.017	
L80	0.655	0.660	+0.005	2/4	+0.022	
L90	0.663	0.668	+0.005	3/4	+0.019	
<i>RegNetY-400MF</i>						
L10	0.178	0.178	+0.000	0/4	-0.012	
L20	0.317	0.317	+0.000	1/4	+0.004	
L30	0.421	0.424	+0.003	3/4	+0.005	
L40	0.502	0.500	-0.002	0/4	+0.006	
L50	0.545	0.543	-0.002	1/4	+0.004	
L60	0.598	0.588	-0.010	0/4	+0.010	
L70	0.623	0.612	-0.011	1/4	+0.008	
L80	0.640	0.637	-0.003	1/4	+0.011	
L90	0.649	0.647	-0.002	1/4	+0.012	
<i>ViT-B/16</i>						
L10	0.169	0.165	-0.004	0/4	+0.021	
L20	0.300	0.293	-0.007	0/4	+0.029	
L30	0.388	0.386	-0.002	0/4	+0.030	
L40	0.460	0.460	+0.000	1/4	+0.034	
L50	0.510	0.507	-0.003	0/4	+0.036	
L60	0.546	0.546	+0.000	1/4	+0.035	
L70	0.575	0.575	+0.000	2/4	+0.028	
L80	0.610	0.604	-0.006	0/4	+0.037	
L90	0.606	0.612	+0.006	2/4	+0.043	

Table S10: **OctMNIST: per-cell constrained-class and overall F1.** One row per (backbone, cap), mean over seeds 1–4. cc-F1 columns compare TraLO to the best constraint-trained baseline (per-seed best of Fioretto-LDF/Hounie-RCL, paired); *win* is the seed-winrate. Δ mac is TraLO minus the best post-hoc clipper on macro-F1. **Bold** Δ marks a paired win ($\Delta \geq 0.005$ on $\geq 3/4$ seeds).

Cap	cc-F1 vs. best trained				macro
	best	TraLO	Δ	win	Δ mac
<i>MobileNetV3</i>					
L10	0.176	0.178	+0.002	1/4	+0.037
L20	0.312	0.312	+0.000	1/4	+0.044
L30	0.391	0.407	+0.016	4/4	+0.036
L40	0.487	0.509	+0.022	4/4	+0.053
L50	0.581	0.583	+0.001	2/4	+0.050
L60	0.635	0.639	+0.004	2/4	+0.049
L70	0.658	0.650	-0.008	1/4	+0.041
L80	0.676	0.677	+0.001	1/4	+0.042
L90	0.689	0.692	+0.002	1/4	+0.045
<i>RegNetY-400MF</i>					
L10	0.171	0.173	+0.002	1/4	-0.001
L20	0.308	0.308	+0.000	1/4	+0.007
L30	0.372	0.407	+0.035	4/4	+0.006
L40	0.452	0.473	+0.021	4/4	+0.006
L50	0.545	0.544	-0.001	1/4	+0.011
L60	0.583	0.603	+0.020	3/4	+0.016
L70	0.626	0.645	+0.019	4/4	+0.018
L80	0.647	0.652	+0.006	3/4	+0.014
L90	0.672	0.671	-0.001	1/4	+0.018
<i>ViT-B/16</i>					
L10	0.176	0.175	-0.002	0/4	+0.000
L20	0.313	0.318	+0.005	2/4	+0.003
L30	0.341	0.422	+0.081	4/4	+0.004
L40	0.462	0.512	+0.051	4/4	+0.005
L50	0.592	0.597	+0.005	4/4	+0.010
L60	0.638	0.643	+0.005	3/4	+0.007
L70	0.671	0.672	+0.001	1/4	+0.010
L80	0.697	0.701	+0.004	2/4	+0.013
L90	0.701	0.714	+0.013	4/4	+0.011

Table S11: **Asymmetric caps (MobileNetV3): per-cell F1.** Global cap L and group cap G set independently; one row per cap pair, mean over seeds 1–4. Columns as in Table S9. The global-tighter ($G < L$) block is where the single preliminary cc-F1 win sits; Δ mac stays positive throughout.

Cap (L/G)	cc-F1 vs. best trained				macro
	best	TraLO	Δ	win	Δ mac
<i>DermMNIST, $G < L$ (global tighter)</i>					
30/20	0.296	0.313	+0.017	3/4	-0.001
50/20	0.295	0.314	+0.019	3/4	+0.029
50/30	0.411	0.417	+0.006	2/4	+0.002
70/20	0.296	0.306	+0.009	2/4	+0.006
70/30	0.413	0.417	+0.005	2/4	+0.006
70/50	0.554	0.565	+0.011	4/4	+0.017
80/20	0.295	0.287	-0.007	1/4	+0.003
80/30	0.411	0.410	-0.001	1/4	+0.033
80/50	0.553	0.563	+0.010	4/4	+0.015
80/70	0.639	0.650	+0.010	4/4	+0.024
<i>DermMNIST, $G > L$ (local tighter)</i>					
20/30	0.312	0.312	+0.000	1/4	+0.003
20/50	0.312	0.310	-0.002	1/4	+0.028
20/70	0.312	0.310	-0.002	0/4	+0.001
20/80	0.312	0.310	-0.002	0/4	+0.003
30/50	0.416	0.417	+0.002	2/4	+0.012
30/70	0.417	0.416	-0.002	1/4	+0.010
30/80	0.417	0.417	+0.000	1/4	+0.018
50/70	0.567	0.561	-0.006	0/4	+0.016
50/80	0.564	0.558	-0.006	0/4	+0.018
70/80	0.645	0.646	+0.001	3/4	+0.025
<i>TissueMNIST, $G < L$ (global tighter)</i>					
30/20	0.171	0.172	+0.001	1/4	+0.015
50/20	0.177	0.166	-0.011	0/4	+0.023
50/30	0.235	0.249	+0.014	3/4	+0.015
70/20	0.179	0.166	-0.014	0/4	+0.014
70/30	0.236	0.216	-0.021	0/4	+0.011
70/50	0.297	0.297	+0.000	2/4	+0.016
80/20	0.179	0.164	-0.015	0/4	+0.013
80/30	0.238	0.245	+0.007	3/4	+0.026
80/50	0.298	0.292	-0.006	2/4	+0.012
80/70	0.324	0.329	+0.005	2/4	+0.016
<i>TissueMNIST, $G > L$ (local tighter)</i>					
20/30	0.188	0.185	-0.002	0/4	+0.011
20/50	0.190	0.178	-0.012	0/4	+0.024
20/70	0.188	0.185	-0.002	1/4	+0.012
20/80	0.188	0.183	-0.005	1/4	+0.008
30/50	0.243	0.243	+0.000	2/4	+0.018
30/70	0.243	0.241	-0.002	1/4	+0.016
30/80	0.243	0.243	+0.000	1/4	+0.026
50/70	0.303	0.317	+0.014	3/4	+0.019
50/80	0.298	0.313	+0.016	3/4	+0.016
70/80	0.340	0.345	+0.005	2/4	+0.018

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